

Ellis Cain

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Education

University of California, Merced

2021 – 2026 (Projected)

PhD candidate, Cognitive and Information Sciences

- Committee: *Dr. Rachel Ryskin (Chair), Dr. Paul Smaldino, Dr. Michael Spivey*
- Dissertation: *Interactions between aging and language evolution*

University of California, Merced

2025

Master of Science, Cognitive and Information Sciences

- Committee: *Dr. Rachel Ryskin (Chair), Dr. Paul Smaldino, Dr. Michael Spivey*
- Capstone: *Development in the idiolect, aging, and language evolution*

Indiana University

2017 – 2021

Bachelor of Science in Cognitive Science with Highest Distinction

Bachelor of Arts in East Asian Languages and Culture with Highest Distinction

- *Member of Phi Beta Kappa*

Publications

Journal articles:

1. **Cain, E.**, & Ryskin, R. (Forthcoming in OpenMind). Semantic Representations are Updated Across the Lifespan Reflecting Diachronic Language Change. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4883002 [🔗](#)
2. **Cain, E.**, Ryskin, R., & Yu, C. (2025). Cross-Situational Statistics Present in an Early Language Learning Context: Evidence from Naturalistic Parent-Child Interactions. *Cognitive Science*, 49(6), e70078. <https://escholarship.org/uc/item/3vq7v1s8> [🔗](#)

Preprints:

1. **Cain, E.**, Geballe, E., & Marghetis, T. (Preprint). A literary translation cycle disentangles transmission pressures from sociocultural changes in cultural evolution. https://doi.org/10.31234/osf.io/9kvb4_v2 [🔗](#)

Textbook chapters:

1. **Cain, E.**, & Yoshimi, J. (2024). Word embeddings. In J. Yoshimi (Ed.). *Neural Networks for Cognitive Science*. <https://downloads.jeffyoshimi.net/NeuralNetworksCogsci.pdf> [🔗](#)

Conference proceedings:

1. Chao, A., **Cain, E.**, & Ryskin, R. (2025). Language experience and prediction across the lifespan: evidence from diachronic fine-tuning of language models. In *Proceedings of the Annual Meeting of the Cognitive Science Society (Vol. 47)*. <https://escholarship.org/content/qt83b1p2nz/qt83b1p2nz.pdf> [🔗](#)
2. **Cain, E.**, & Ryskin, R. (2024). Diachronic change in verb usage statistics predicts differences in sentence processing across the lifespan. In *Proceedings of the Workshop on Cognitive Modeling and Computational Linguistics* (pp. 221-230). <https://aclanthology.org/2024.cmcl-1.19.pdf> [🔗](#)
3. **Cain, E.**, & Ryskin, R. (2023). Diachronic Language Change and Its Influence on Lexico-semantic Representations Across the Lifespan. In *Proceedings of the Annual Meeting of the Cognitive Science Society (Vol. 45, No. 45)*. <https://escholarship.org/uc/item/943334rm> [🔗](#)
4. **Cain, E.**, Ryskin, R., & Yu, C. (2022). Quantifying cross-situational statistics during parent-child toy play. In *Proceedings of the Annual Meeting of the Cognitive Science Society (Vol. 44, No. 44)*. <https://escholarship.org/uc/item/7w66j9bd> [🔗](#)

5. Zhang, Y., **Cain, E.**, Crandall, D., & Yu, C. (2022). Grounding Action Verbs in Egocentric Visual Perception. *In Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 44, No. 44). <https://doi.org/10.31234/osf.io/yzf63> 
6. Zhang, Y., Amatuni, A., **Cain, E.**, Wang, X., Crandall, D., & Yu, C. (2021). Human Learners Integrate Visual and Linguistic Information Cross-Situational Verb Learning. *In Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 43, No. 43). <https://escholarship.org/uc/item/6gt5t65g> 
7. Zhang, Y., Amatuni, A., **Cain, E.**, & Yu, C. (2020). Seeking Meaning: Examining a Cross-situational Solution to Learn Action Verbs Using Human Simulation Paradigm. *In Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 42, No. 42). <https://escholarship.org/uc/item/62f132f7> 

Presentations

Invited talks:

1. Chao, A., **Cain, E.**, & Ryskin, R. (2025). Language experience and prediction across age groups: evidence from diachronic fine-tuning of language models. *For the MRPI Planning Grant: Leveraging California's Linguistic Diversity to Improve Large Language Models*. University of California

Talks:

1. Chao, A., **Cain, E.**, & Ryskin, R. (2025). Language experience and prediction across age groups: evidence from diachronic fine-tuning of language models. *At the California Meeting on Psycholinguistics (CAMP[8])*. Davis, CA
2. **Cain, E.**, Chao, A., & Ryskin, R. (2024). Diachronic language change explains apparent age-related differences in information-theoretic efficiency. *At the Conference on Rational Approaches in Language Science (RAILS)*. Saarbrücken, Germany
3. **Cain, E.**, Chao, A., & Ryskin, R. (2024). Stability in the production of syntactic attachment across the lifespan. *At the California Meeting on Psycholinguistics (CAMP[7])*. San Diego, CA
4. **Cain, E.**, & Ryskin, R. (2024). Diachronic change in verb usage statistics predicts differences in sentence processing across the lifespan. *At the Workshop on Cognitive Modeling and Computational Linguistics*. Bangkok, Thailand
5. **Cain, E.**, Brown-Schmidt, S., & Ryskin, R. (2023). Diachronic Verb Usage Statistics and Sentence Processing Across the Lifespan. *At the Annual Meeting of the Psychonomic Society*. San Francisco, CA
6. **Cain, E.**, Ryskin, R., & Yu, C. (2022). Quantifying cross-situational statistics during parent-child toy play. *At the Annual Meeting of the Cognitive Science Society*. Toronto, Canada

Posters:

1. Chao, A., **Cain, E.**, & Ryskin, R. (2025). Language experience and prediction across the lifespan: evidence from diachronic fine-tuning of language models. *At the Annual Meeting of the Cognitive Science Society*. San Francisco, CA
2. **Cain, E.**, Chao, A., & Ryskin, R. (2024). Diachronic language change explains apparent age-related differences in information-theoretic efficiency. *At the California Meeting on Psycholinguistics (CAMP[7])*. San Diego, CA
3. **Cain, E.**, Geballe, E., & Marghetis, T. (2024). Translation as Transmission: Disentangling the impact of cultural context from cultural transmission using an unusual English-to-English cycle of 19th century literary translations. *At the Cultural Evolution Society Conference (CES)*. Durham, UK
4. **Cain, E.**, & Ryskin, R. (2024). Neural Signatures of Language Processing Across the Lifespan. *At the Symposium of the Center for Interdisciplinary Neuroscience*. Merced, CA
5. **Cain, E.**, & Ryskin, R. (2023). Diachronic Language Change and Its Influence on Lexico-semantic Representations Across the Lifespan. *At the Annual Meeting of the Cognitive Science Society*. Sydney, AUS

6. **Cain, E.** (2023). Teaching An Old Dog New Tricks? Learning Rates, Aging, And Language Change. *At California Workshop on Evolutionary Social Science (CWESS)*. Merced, CA
7. **Cain, E., & Ryskin, R.** (2022). Diachronic language change and meaning across the lifespan. *At Architectures and Mechanisms of Language Processing (AMLaP)*. York, UK

Research experience

Language, Interaction, and Cognition Lab

2021 – Present

PI: Dr. Rachel Ryskin; *University of California, Merced*

- Designed, implemented, and ran behavioral experiments using JsPsych and Prolific
- Designed, implemented, and ran electro-encephalography (EEG) experiments using PsychoPy and EEGLAB
- Recruited and worked with older adult participant populations
- Conducted large-scale corpora analyses and developed LLM-based annotation tools
- Helped manage our online experiment server using JATOS
- Trained and mentored 5 undergraduate research assistants and a lab manager on running EEG experiments and data pre-processing, running online experiments through Prolific, and basic statistical analysis using R

Developmental Intelligence Lab

2020 – 2022

PI: Dr. Chen Yu; *University of Texas, Austin*

- Worked on a project exploring the ecological validity of cross-situational learning using semi-naturalistic toy play data
- Created a data-processing pipeline to extract label-object co-occurrences from multi-modal data-streams

Cognitive Development Lab

Summer 2020

PI: Dr. Linda Smith; *Indiana University*

- Worked with a post-doctoral researcher to adapt an Eulerian magnification script to process previously-collected video and extract heartbeats from an eye-camera data-stream

Computational Cognition and Learning Lab

2019 – 2020

PI: Dr. Chen Yu; *Indiana University*

- Worked with two graduate students to collect cross-situational verb learning data through an online experiment
- Compared and implemented various word similarity calculation methods (i.e., WordNet, word embeddings) to evaluate free response data from the online experiment

Teaching Experience

Teaching assistant – Language and Linguistics

Fall 2025

IoR: Dr. Zenaida Aguirre-Muñoz; *University of California, Merced*

- Led weekly discussions sections to help students engage with the lecture materials
- Collaborated with Dr. Aguirre-Muñoz with designing methods to teach students responsible AI usage in learning

Guest lecture – Introduction to Language and Linguistics

Fall 2022; Fall 2025

IoR: Dr. Zenaida Aguirre-Muñoz; *University of California, Merced*

- Presented guest lecture on statistical word learning in first-language acquisition, with a special focus on the ecological validity of the proposed theories

Teaching assistant – Research Methods in Psychology

Fall 2024

IoR: Dr. Yueqi Yan; *University of California, Merced*

- Introduction to the methods of psychological research, covering from hypothesis development to data visualization and basic statistical analysis
- Led weekly laboratory sections to answer questions and review lecture concepts

Guest lecture – Research Methods in Psychology

Fall 2024

IoR: Dr. Yueqi Yan; *University of California, Merced*

- Guest lecture on data collection and visualization, including a live demonstration of online data collection and graphing using Excel and R/GGplot

Teaching assistant – Language in Minds, Brains, and Machines

Spring 2024

IoR: Dr. Rachel Ryskin; *University of California, Merced*

- Collaborated with Dr. Ryskin and Dr. Jeff Yoshimi to develop course materials
- Co-authored the textbook chapter on Word Embeddings in Dr. Yoshimi's Neural Networks textbook and designed the accompanying evaluation activities
- Created a demonstrative teaching tool for word embeddings in Dr. Yoshimi's SimBrain
- Provided in-class aid with the programming activities
- Provided support for final group projects, where students conducted a replication study and analyzed their own data using R and Python

Guest lecture – Language in Minds, Brains, and Machines

Spring 2024

IoR: Dr. Rachel Ryskin; *University of California, Merced*

- Presented guest lecture on lexical semantics, with a special focus on the overlap between distributional semantics and statistical learning
- Integrated original and existing research

Guest talk – Foundations of Cognitive Science

Fall 2023

IoR: Dr. Tyler Marghetis; *University of California, Merced*

- Presented guest lecture on language change and lifespan semantic representations, with an additional focus on data visualization and presentation skills

Teaching assistant – Introduction to Cognitive Science

2021 – 2022

IoR: Dr. Michael Spivey; *University of California, Merced*

- Introduction to Cognitive Science, surveying the perspectives and methods from the various comprising fields
- Led weekly discussion sections to review and reinforce lecture materials
- Created discussion and review activities

Grants, Fellowships, and Workshops

Graduate Dean Dissertation Dean Fellowship

Spring 2026

MacKenzie Scott Graduate Student Supplemental Travel Award

2023

Pushkin Hackathon, located at Boston College

2023

Languages

Natural: English (native), Mandarin Chinese (advanced)

Computer: *Expert:* R, Python, MATLAB, LaTeX

Advanced: Julia, NetLogo

Intermediate: Java, Javascript, Kotlin